TEACHING PHILOSOPHY STATEMENT

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My teaching philosophy is centered on the fact that learning is a human endeavor. First, as a mathematics instructor, it is crucial to frame problem solving around storytelling in logic rather than rote memorization or numerical solutions. This approach helps students from diverse backgrounds develop conceptual understanding that will last beyond the end of the course. Second, as an educator more broadly, my aim is to cultivate an environment where all students are comfortable and know that I recognize their lives beyond the classroom. The most effective way to encourage independent mathematical thinking is to reveal the possibilities for exploration inherent in mathematics while acknowledging the humanity of both the subject and the student.

1. ENCOURAGING EXPLORATION

I enjoy creating course materials that invite students to explore mathematics, treating explanations as opportunities for both logical argument and creative expression. For example, as an instructor of record for Calculus for Business and Economics, I developed a sequence of six interactive Data Science Demos using Streamlit to extend the course material beyond traditional content. Each demo centered on a real-world scenario through which students could explore applications of introductory calculus to data science. In one, students investigated cost function optimization and gradient descent by fitting a linear model to mock airline pricing data; in another, they explored classification and logistic regression by predicting whether or not a Netflix subscription would remain active. These demos allowed me to develop engaging visualizations and connect course concepts to applications relevant to business and economics majors, while presenting mathematics as something to investigate rather than simply a set of procedures to follow. Students echoed these goals in their feedback, describing the demos as helping them "conceptualize concepts in math" and noting that, "content-wise they were enjoyable."

In a similar vein, as a teaching assistant I designed collaborative activities for honors discussion sections of Calculus II and Multivariable Calculus, using guided worksheets, visualizations, and conceptual notes to facilitate student-driven coverage of advanced topics. For example, one Calculus II activity guided students through a proof of the irrationality of e using the remainder in its series expansion, while a Multivariable Calculus activity introduced continuous probability and expected values through a dart game, combining interactive simulations with the mathematical development of probability densities. In both cases, I sought to build a structure that would support rigor without prescribing every step of the discussion. Rather than simply lecturing, I encouraged students to shape the conversation through their questions and observations, adapting my explanations to their thinking as we progressed. One student described me as "great at understanding how we learn and adjusting on the fly."

2. PUTTING PERSONALITY IN PEDAGOGY

In my experience, students are more willing to engage with mathematics when they know their instructor sees them as people, not just learners. While our collective goal is to highlight

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the plot that unfolds between the lines of proofs or behind the numbers of calculations, I believe students are better equipped to pursue that goal when the classroom leaves room for life beyond mathematics. To that end, I use lighthearted humor and conversational dialogue during natural lecture pauses to reduce the tension of formality. I also bring personality to my written materials, whether through an Album of the Week in my lecture notes or the occasional pop-culture reference in an interactive demo. By making room for points of connection beyond the mathematics itself, I hope to make the subject feel less imposing and the classroom a place where students can delve into it more comfortably. When asked in an end-of-semester survey to describe positive aspects of my teaching, a student replied "I liked the album of the week, it made it more personable." Five other responses noted that the weekly album gave them something to look forward to in association with math. Such choices may seem small, but they carry significant pedagogical weight; as expressed by a student, "...[Cole's] positive attitude and good humor are literally what got me out of bed on Monday and Wednesday mornings. He went over example problems very clearly and made difficult concepts easier to understand. Cole made me forget how much I hate calculus and couldn't think of anyone else better to TA this class."

3. Leadership

I believe effective teaching extends beyond the individual classroom: it also requires collaboration among instructors and thoughtful decisions about how courses are structured. As an instructor of record, I have taken responsibility for the design and implementation of my courses, including assessment policies intended to encourage students to learn from mistakes and demonstrate their growth. In Elementary Differential Equations, for example, students could earn back half the points they lost on quizzes through corrections, and I replaced a student's midterm grade with their final exam grade whenever the latter was higher. These policies reflect my view that assessment should not merely measure what students know at a single moment, but also give them opportunities to demonstrate improvement.

I have also taken on a leadership role with responsibilities beyond my own discussion sections. As the department's first Lead TA for Multivariable Calculus, I helped coordinate a team of graduate instructors. My responsibilities included organizing shared office hours, preparing weekly exam-proctoring schedules, communicating with instructors, and arranging substitute coverage. This required balancing the needs and availability of my colleagues with the department's obligation to provide students with consistent instructional support. In this role, I aimed to foster open communication across the course while helping my fellow TAs navigate the practical challenges of teaching. My experiences working both independently as an instructor and collaboratively within a teaching team have shown me that good teaching depends not only on what happens in the classroom, but also on the structures and relationships that make it possible.

I view teaching as a continuous process of reflection and refinement. I regularly revise my lecture notes to improve their explanatory precision and adapt my pedagogical choices in response to student feedback, always seeking better ways to make mathematics both accessible and engaging. I look forward to bringing this same commitment to future courses and collaborations, learning from colleagues while refining my approach to teaching. Ultimately, I aim to help students not only understand mathematics, but also approach it with greater curiosity and confidence.